

Upgrading Energy Technology—A Free-Market Solution

While politicians were debating how many billions in tax breaks should go to the energy industry and how much should go to renewable resources, home inspectors have been uncovering an untapped energy resource available in millions of American homes—replacing outdated energy technology. If all homes in the United States acquired today's energy technology, we would use 25% less energy in our homes, saving about \$30 billion per year. These technologies are mass-produced, widely available, and competitively priced. They need no tax incentives, though they may cost somewhat more initially. The

energy savings resulting from these technologies more than compensate for the additional cost. I can say this after many years of experience directing an energy audit company that I founded, CMC Energy Services.

Significant improvements have been made in residential energy technology during the past two decades. Today's air conditioners, refrigerators, and heat pumps use half as much electricity as those made 20 years ago. Compact fluorescent lights use one-third the electricity required for incandescent lights. Today's thermal low-e windows lose much less heat in the winter and keep out more heat from the sun in the summer than yesteryear's windows. Building codes now require more insulation than ever before. And thanks to today's

detergents, savvy consumers launder with cold water, using only 20% of the energy they previously used for washing clothes.

Some recent events have made energy efficiency retrofits even more attractive for homeowners. There is a 10% tax credit for residential energy-efficient technologies in the 2005 Energy Bill, to be available in 2006 and 2007. The new Federal Housing Authority (FHA) Streamlined 203(K) Limited Repair program, which allows the addition of \$5,000–\$15,000 for repairs to an FHA mortgage, includes energy efficiency improvements.

The major hurdle to overcome is selling energy upgrades to homeowners. There are so many homes, all somewhat different, and compared to the savings for commercial buildings, the energy savings

Home Energy Tune-uP

A number of home inspectors and home inspection companies have already added an energy inspection to the services they provide their clients. Home Energy Tune-uP has trained and certified 400 energy inspectors in 39 states in the past year. The program is flexible in terms of application. An energy inspection can be done in conjunction with a general home inspection or separately. Unlike a general home inspection, which looks for things that are wrong with the house, the energy inspection looks for potential energy improvements. The inspector prints an Energy Improvement Report that lists recommendations and estimated savings and costs. In addition, the report lists those technologies that are eligible for special financing because their savings exceed their costs. The report discusses each recommendation and explains additional low-cost techniques for saving energy. A team of technical experts offers help with implementation (via phone), and currently available contractors are listed. Home buyers inevitably see that investing in

energy technology improvements not only will make their home more comfortable and more valuable, but will also result in a small positive cash flow as energy prices go up.

The following example shows how Home Energy Tune-uP has worked for Michael Bate, a Realtor in Bethesda, Maryland. Bate was concerned about the high energy bills for his 50-year-old split-level home. There had been an 18% increase in energy prices last year. A local home inspection company, Faro Systems, Incorporated, had told him about Home Energy Tune-uP. Camilo Galeano, a Faro Systems inspector, examined Bate's home and noted his findings on the Tune-uP data collection form. These data were then entered into a Web-based software program that generates the report.

Table. Improvements Eligible for Special Financing

Measures	Recommendation	Annual Savings	Cost
Air seal-up	Seal up	\$273	\$ 300
Water heater	Consider replacing—Age	\$ 71	\$ 790
Heating system # 2	Consider replacing—Age	\$152	\$2,160
Floor #2	Insulate to R-25	\$ 16	\$ 243
Windows	Replace w/double-e	\$ 57	\$3,355
Total		\$569	\$6,848
Monthly savings and cost + interest		\$ 47	\$ 41

The house has two heating systems. Galeano noted that, while both systems worked well, one was 33 years old and had a design life of only 20 years. The Tune-uP report recommended: "Consider Replacing—Age." This was also the recommendation for the water heater, which was 21 years old and had a design life of 11 years. Why recommend replacement for something that still works? First, because some of today's energy technology is far more efficient than that of 15 years ago, and replacing just those two items would save the homeowner \$223 a year—more than 10% of his energy bill. Second, it is far better to plan major pur-

for these homes are relatively small—\$450 per year on average, based on today's energy prices. However, a window of opportunity opens up when someone buys a house—and there were 7 million such opportunities last year! This window of opportunity, the point of sale, is when the best financing terms are available to the buyer, and when most home improvement plans are made. It is also the time when inspectors perform a general home inspection and could provide an energy inspection. (For an example of a successful home energy inspection program supported by my company, see "Home Energy Tune-up.")

Two key attributes give energy efficiency upgrades an edge over other home improvements. First, they create a stream of savings over time—the savings on the monthly energy bill. Second, they add to a home's value. We estimate that a \$7,000 investment in energy upgrades will increase the value of a

home by approximately \$10,000. These two factors make energy efficiency improvements an excellent investment, not only for the home buyer, but also for the mortgage lender. Smart mortgage lenders will finance cost-effective energy improvements more readily than other improvements, since they don't have to re-qualify the borrower or the home. Also, by replacing costly equipment, such as a heating system, water heater, or air-conditioner before it fails, lenders and home buyers minimize future large unexpected costs that threaten timely mortgage payments.

For homeowners, the welcome benefits of the energy inspection program are a more comfortable home, a more valuable home, and savings in energy costs. Even a homebuyer with limited means can invest in energy improvements, since financing is available for those improvements that cost less than they save.

For other sectors of the economy, energy efficiency upgrades also create significant benefits. Home inspectors who add energy efficiency analyses to their offerings increase their income and add work during slow periods. And just as the inspection industry boomed when Realtors realized that an extra pair of trained eyes would protect them from future liability, so clients will prefer to do business with a Realtor whose inspectors can help them do an energy fix-up on their newly purchased home at no out-of-pocket cost to them. The largest overall economic payoff would be for contractors, including suppliers and installers of insulation, window replacement, heating and cooling equipment, and appliances and weatherization.

Finally, while the Energy Policy Act of 2005 emphasizes the supply side of the energy equation, consumers still have the final say when it comes to demand. An energy inspection offers the public a clear choice. Acquire today's energy technology for yesteryear's homes financed by tomorrow's energy savings, or do nothing and keep using outdated energy technology, with the inevitable result of a less comfortable home, increased air pollution, higher energy bills, more dependence on foreign fuel, and a lower resale value of the house.



—Doris Iklé

Doris Iklé founded and directs CMC Energy Services, which is headquartered in Bethesda, Maryland, and has offices in Fort Washington, Pennsylvania, and Princeton Junction, New Jersey.

chases and take time to compare quality, prices, and financing than to wait until things break down—when Murphy's Law inevitably steps in.

In the living room, the home inspector made note of the fact that the large picture window had not been upgraded like the other windows. The Tune-up report recommended that this window be replaced with a double-pane low-e window. Despite the high cost, Bate made this investment because the area around the window always seemed cold in the winter and the glass added unwanted heat in the summer. When asked later which improvement he thought was the most valuable, he immediately said, "The window," because it made a real difference in the comfort of his home. Also, as a Realtor, Bate knew that replacing this window can increase the resale value of the house by more than the cost.

Other recommendations that Bate implemented were to put a ridge vent on the roof to increase air circulation, clean out the vents, and replace a single-glazed window in the basement that was in need of repair. The report also recom-

mended insulating the floor over the crawlspace.

Although Bate had not planned to finance these improvements, he was surprised when the report showed that if he did finance them as part of a mortgage, the monthly cost (including interest) would be less than the estimated savings. The Tune-up report separates both the individual items that save more than they cost when financed, and the optimum group of items that will meet the criteria. The large picture window by itself would never have qualified, but it did qualify when added to the other recommendations. In fact, the monthly savings—\$47—exceeded the monthly cost—\$41, including interest (see Table).

"But what would happen if someone moved out after just one year?" Bate asked. With a 30-year loan, they would have paid only one-thirtieth of the cost and obtained the energy savings for just one year, but the resale value would have increased by at least the full cost of the improvements recommended by Home Energy Tune-up.

FOR MORE INFORMATION:

For more on the FHA Streamlined 203(K) Limited Repair program, go to www.hud.gov/offices/hsg/sfh/203k/203kslrp.cfm. For more information about Home Energy Tune-up, go to www.hometuneup.com.